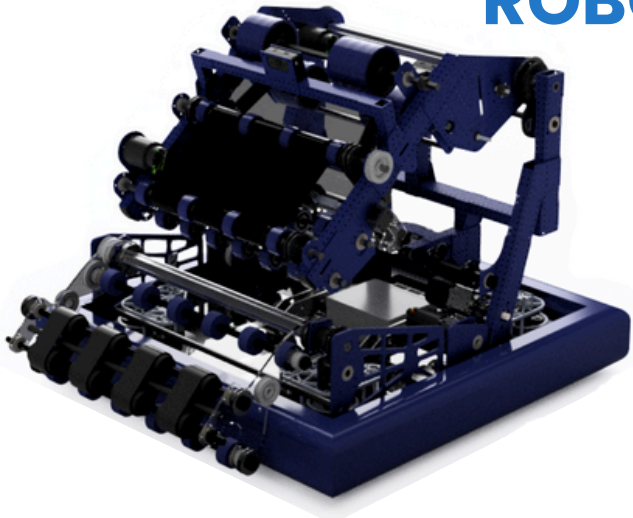


TEAM 687

THE NERD HERD

ABRUTICUS

ROBOT OVERVIEW



SHOOTER

Flywheel:

- Vertical Orientation
- 4in. stealth wheels

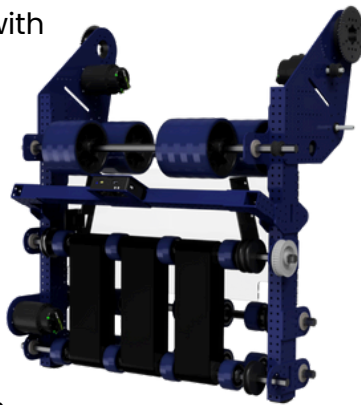
Indexer:

- 3 Polyurethane belts with crowned pulleys for smoother indexing

Scoring Location

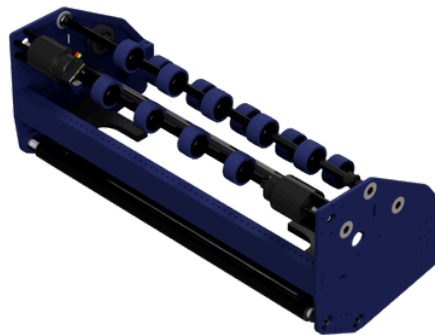
Capabilities:

- Speaker
- Stage
- Amp
- Trap (In Development)



- 27 in by 27 in
- Four MK4i Swerve Modules L1 (equipped with 2 Krakens per module)
- Black Neoprene Tread Wheels 1 x 1 x 1/16 in
- Handrails Chassis: 2x1x1/16 Aluminum Tubes

DRIVE BASE



INTAKE

Rollers

- 1.5 in OD VersaRollers
- 2 in Stealth Wheels
- GT2 belts and pulleys
- T 5mm HTD belt

Belt Rigging

- 2 Falcon 500s each driving one direction
- 2:1 ratio from Falcon 500s to VersaRollers

Capabilities:

- Pick up notes
 - Ground

AFRAME

- 142 deg/s, Stow to Amp shot takes .64 seconds
- Double motor setup allows for speed, while having enough torque to climb using the shooter as an arm, as a backup option

Parts:

- MAX Spline rotating on bushing on .75" Tube dead axle
- 1:1 ratio 3D-printed gears rotate a absolute encoder
- Powered by 2 Falcon500 Motors with 75:1 MAX Planetary gearbox, and a 3:1 chain drive



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HAAS
Gene Haas Foundation



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THE NERD HERD

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DESIGN PROCESS AND BUILD SEASON OVERVIEW OFFSEASON

- Analyzed 2023 Charged Up season (what worked and what didn't work during the season?)
- 3 Offseason robots and preparation for Chezy Champs
- Ran an offseason Competition SoCal Showdown
- Held summer workshops for returning members
- Worked on better understanding the design of common subsystems like Claw, elevator, shooter

WEEK 1

- Kickoff and team discussion on possible robot archetypes
- Built game arena and KitBot
- Researching and drafting our first iterations of subsystems
- First Iterations where being approved for Fabrication

WEEK 2

- Further prototyping
- Continue drafting/approvals of first iterations of subsystems
- Manufacturing

WEEK 3

- Manufacturing
- Staggered subsystem Assembly of first iteration of Robot
- Parts of Alpha bot being assembled

WEEK 4

- First iteration robot is being tested
- Autos being implemented and tuned
- Version 2 designs approved and getting manufactured
- Prepping robot for powder coat

WEEK 5

- Competition Robot Finalized
- Subsystem spares are being made
- Climb endgame + trap being designed & tested

WEEK 6 AND BEYOND

Trap testing
MORE TESTING

At the beginning and end of each meeting our team makes it a priority to do a team brief

Pre-brief: goals each subteam wants to accomplish by the end of the meeting

Debrief: Overview of the actual progress the subteam made

Leadership was constantly at 4201 The Vitruvian Bots practice field

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Gator House Foundation

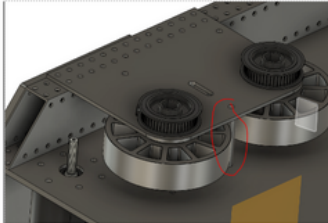




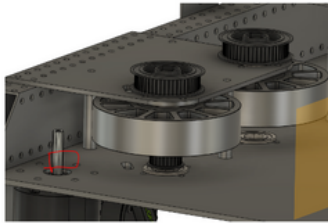
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THE NERD HERD ABRUTICUS

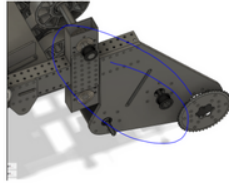
OVERVIEW OF DESIGN PROCESS



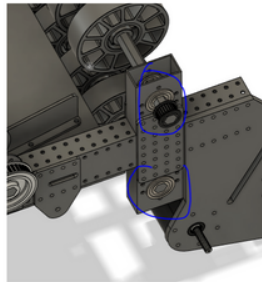
Move this standoff. The wheels will expand as it shoots



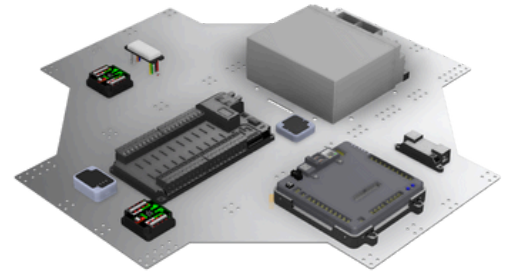
Add the pulley with proper hardware in



I think we need to find a way to reuse the current pivot plates



Change the 2x1 to 1/8 aluminum plate instead



Product of Design review
w/ Assembly

JA You
Jan 22, 7:59 AM

Praise



Growth
Phong Trinh

You really cooked on intake. You are a great asset to Nerd herd 687.

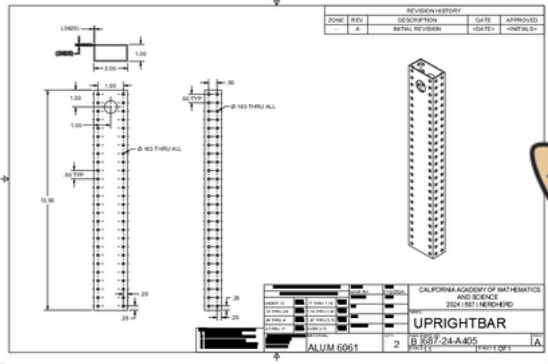
From Janet Favor Ani

See more

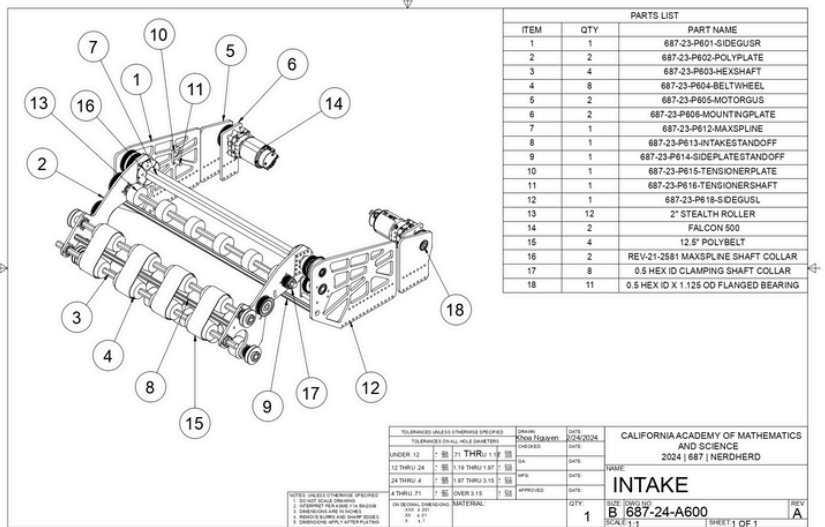
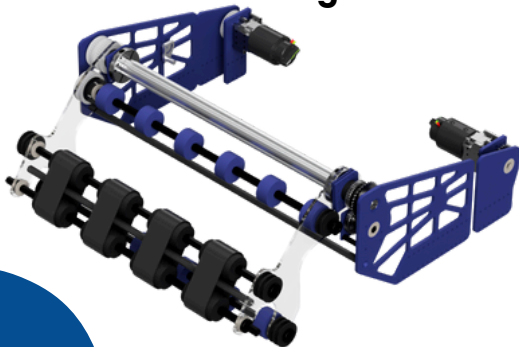
Give praise to design members
important step

January 11 design review

January 26 Design Review



Technical Drawing approved for
Manufacturing

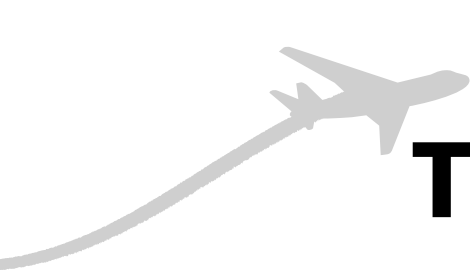


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SM

Finalized Design and Assembly Drawing





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LINKS TO DOCUMENTATION



Technical Binder



Software Binder



Outreach Documentation



Business Plan



Manufacturing Documentation



Safety Manual

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THE NERD HERD ABRUTICUS

OFFSEASON

Group 2

Christopher Morales &
Christopher Ramirez Manzo

Plus

- Documentation
- Revision of work/improvement of product
- Awards
 - LAR, Dean's List, Archimedes
- Prototypes
- Field Elements, Quick Assembly
- Quick Manufacturing

Delta

- FRC: Organization
- Distribution of work
- Communication Between Sub Teams
- Assembly held on to robot to long, not enough time for programming
- Interactions between sub teams
- CAD: Not trained on technical stuff
- FTC: Material Organization
- Prog Only Computers

Group 5

Julianna Salonga & Janet Ani

Plus

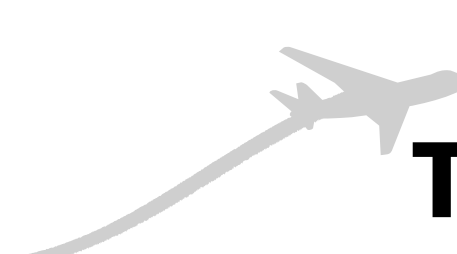
- FTC
 - Regularly scheduled meetings (no confusion)
 - Different people specialized into different "subteams" but still collaborated
 - Frequent field testing
 - Dedicated team members
- FRC
 - Worlds!
 - Ri3d
 - Da Vinci
 - Clear goals and agendas (briefing)
 - Distribution of work amongst members
 - Mentorship
 - People participated in scouting

Delta

- FTC
 - Lack of clear due dates/agendas
 - FTC and FRC skills are different (training didn't match with FTC needs)
 - Quality of solutions (solutions applied were always quick and temp.)
 - Lack of organization of parts throughout season
- FRC
 - Disorganized (Drawings, Tools, Parts)
 - Didn't keep up with Open Alliance
 - Sportsmanship (have gp please)
 - Late to meetings
 - Lack of spares
 - Communication
 - Design for Prog too "heart"
 - Didn't prioritize documentation
 - Have not posted on YouTube in 6 years :(

Separated team into 10 groups analyzing
pluses and deltas of 2023 season

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OFFSEASON

AREAS OF IMPROVEMENT

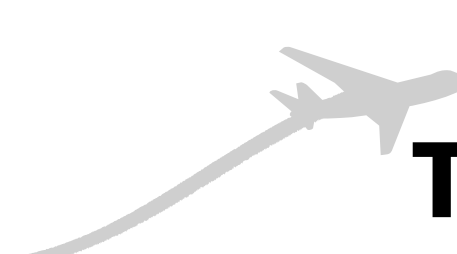
- Communication
- Organization
- Outreach
- FTC Merch/buttons
- Documentation
- Time management (too much downtime)
- Attendance
- Open alliance
- Unequal work distribution
- Dirty Shop
- More FTC mentorship

GOALS

- Functional, high-scoring robot
- More subteam bonding
- More FTC inclusion
- Learn about CAD intakes
- Consistent vision
- Learn more about electrical
- Stronger business plans
- Better outreach
- MFG: members use both machines
- Complete training in all subteams before season starts
- Announce meetings earlier
- Robot finished a week before first comp
- Working scouting app
- IsMe2 for offseason comps



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OFFSEASON

SKILLS

- More cross-training
- Leadership
- Documentation
- Torque calculations
- Path planner & markdown
- Electrical connections/how to make wires
- How to use lathe
- PROG: more detailed planning/organization
- Media
- Time estimation
- FTC: more CAD & 3D printing skills
- Comfortable communication w/ leads
- Technical skills through offseason projects

NEW ROBOT IDEAS

- Swerve drive
- LEDs
- Improve speed --> lower center of gravity
- @ max velocity but acceleration is currently capped
- FTC: mecanum drive
- Ground intake
- Custom swerve (like 4414)
- Faster extension
- Sensors
- Intake from multiple sources
- More autos
- More backup parts (gyros)
- Arm that swings in both directions
- More compact
- Be able to get up after falling



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Socal Showdown

Team 687 The Nerd Herd





SO CAL SHOWDOWN OFFSEASON COMPETITION

Our team's most recent endeavor, co-running an off-season competition called SoCal Showdown, proved to be one of our biggest successes. Near the heart of Los Angeles buzzing with FRC teams, the two closest offseason events are hours away and have a tendency to fill up quickly. In turn, SoCal Showdown was born. The purpose of SoCal Showdown was to create an off-season competition that was accessible to local teams in the Los Angeles area. The event was created to allow local teams to exchange knowledge and gain connections with mentors, in addition to getting a fun, memorable competition experience.

Hosted By Team 687 The Nerd Herd, Team 4201 The Vitruvian Bots, and Team 2637 Phantom Cats

